

Rail Engineering Mathematics

Tutorial Sheet 5

Questions

1. Given the following complex numbers in Cartesian form:

$$z_1 = 7 - j3, \quad z_2 = -1 - j4, \quad z_3 = -5 + j, \quad z_4 = 9 + j6$$

- (i) Express in polar and exponential form:

(a) z_1

(b) z_2

(c) z_3

(d) z_4

- (ii) Calculate the following in polar form:

(a) $z_3 z_1$

(e) $\frac{z_1}{z_3}$

(b) $z_2 z_4$

(f) $z_3 z_2$

(c) $\frac{z_3}{z_1}$

(g) $\frac{1}{z_2}$

(d) $\frac{z_4}{z_2}$

(h) $\frac{1}{z_4}$

2. Given the following complex numbers in Polar form:

$$z_5 = 2.5\angle -2.9, \quad z_6 = 4.1\angle -5.1, \quad z_7 = 0.3\angle 1.7, \quad z_8 = 7.9\angle 6.1$$

(i) Express in rectangular/Cartesian form:

(a) z_5

(b) z_6

(c) z_7

(d) z_8

(ii) Calculate the following in polar form, and then convert the result to Cartesian form:

(a) $z_5 z_6$

(d) $\frac{z_8}{z_5}$

(b) $z_7 z_8$

(e) $\frac{1}{z_7}$

(c) $\frac{z_6}{z_8}$

(f) $\frac{1}{z_5}$

3. When multiple impedances in a electrical circuit are connected in series, the total impedance Z (ohms) is given by the sum of the individual impedances. This is related to the voltage V (volts) and current I (amps) using Ohm's Law, which states that $V = IZ$.

Two impedances $Z_1 = (3 + j6) \Omega$ and $Z_2 = (4 - j3) \Omega$ are connected in series to a supply voltage of 120 V. What is the magnitude of the current flowing through the circuit?